SAVEETHA SCHOOL OF ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

COMPUTER SCIENCE AND ENGINEERING

CSA47 – DEEP LEARNING

ASSIGNMENT – 1

CO1 – MATHEMATICAL FOUNDATIONS AND MACHINE LEARNING PRINCIPLES

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Course:	CSA47 – Deep Learning	Nature:	Individual

Total Marks: 50

CO1: Analyze the foundational mathematical concepts, including linear algebra and probability, and core machine learning principles like bias-variance trade-offs and supervised/unsupervised learning algorithms.

Topics: Linear Algebra, Probability and Information Theory, Numerical Computation, Capacity, Overfitting/Underfitting, and Bayesian Statistics.

Purpose

Investigate how mathematical foundations influence the reliability of learning systems. Students must connect equations, assumptions and numerical behaviour to practical model decisions.

Scenario: Healthcare Risk Prediction

A hospital plans to develop an intelligent system to predict disease risk using patient age, blood pressure, glucose level, cholesterol and family history. The dataset contains labelled patient records as well as unlabelled records. You are required to analyze the data, select suitable learning approaches and justify decisions using mathematical reasoning.

TASK 1 – Patient Data Representation and Normalization (5 Marks)

a) Effect of using raw features without centering or normalization

The hospital collects patient information such as age, blood pressure, glucose level and cholesterol. These features have different numerical ranges.

For example:

Feature	Minimum value	Maximum value
Age	25	60
Blood Pressure	120	160
Glucose Level	90	130
Cholesterol	180	260

If raw features are used without normalization:

1. Features with larger numerical values may dominate the learning process.
2. Distance-based algorithms may calculate misleading similarities.
3. Gradient-based algorithms may converge slowly.
4. Model training may become unstable.
5. Features with smaller scales may receive less influence than intended.

Therefore, normalization is required to bring features into a comparable numerical range.

b) Feature matrix and normalization

Consider the following example of three patients.

Original feature matrix:

Patient	Age	Blood Pressure	Glucose	Cholesterol
P1	25	120	90	180
P2	45	140	130	220
P3	60	160	110	260

The feature matrix is:

$$X = \begin{bmatrix} 25 & 120 & 90 & 180 \\ 45 & 140 & 130 & 220 \\ 60 & 160 & 110 & 260 \end{bmatrix}$$

Min-Max Normalization

Min-Max normalization converts values into a range between 0 and 1.

Formula:

$$X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

Example for Patient 1's age:

$$Age' = \frac{25 - 25}{60 - 25} = 0$$

Example for Patient 2's age:

$$Age' = \frac{45 - 25}{60 - 25} = 0.5714$$

Example for Patient 3's age:

$$Age' = \frac{60 - 25}{60 - 25} = 1$$

Normalized feature matrix:

Patient	Age	Blood Pressure	Glucose	Cholesterol
P1	0.000	0.000	0.000	0.000
P2	0.571	0.500	1.000	0.500
P3	1.000	1.000	0.500	1.000

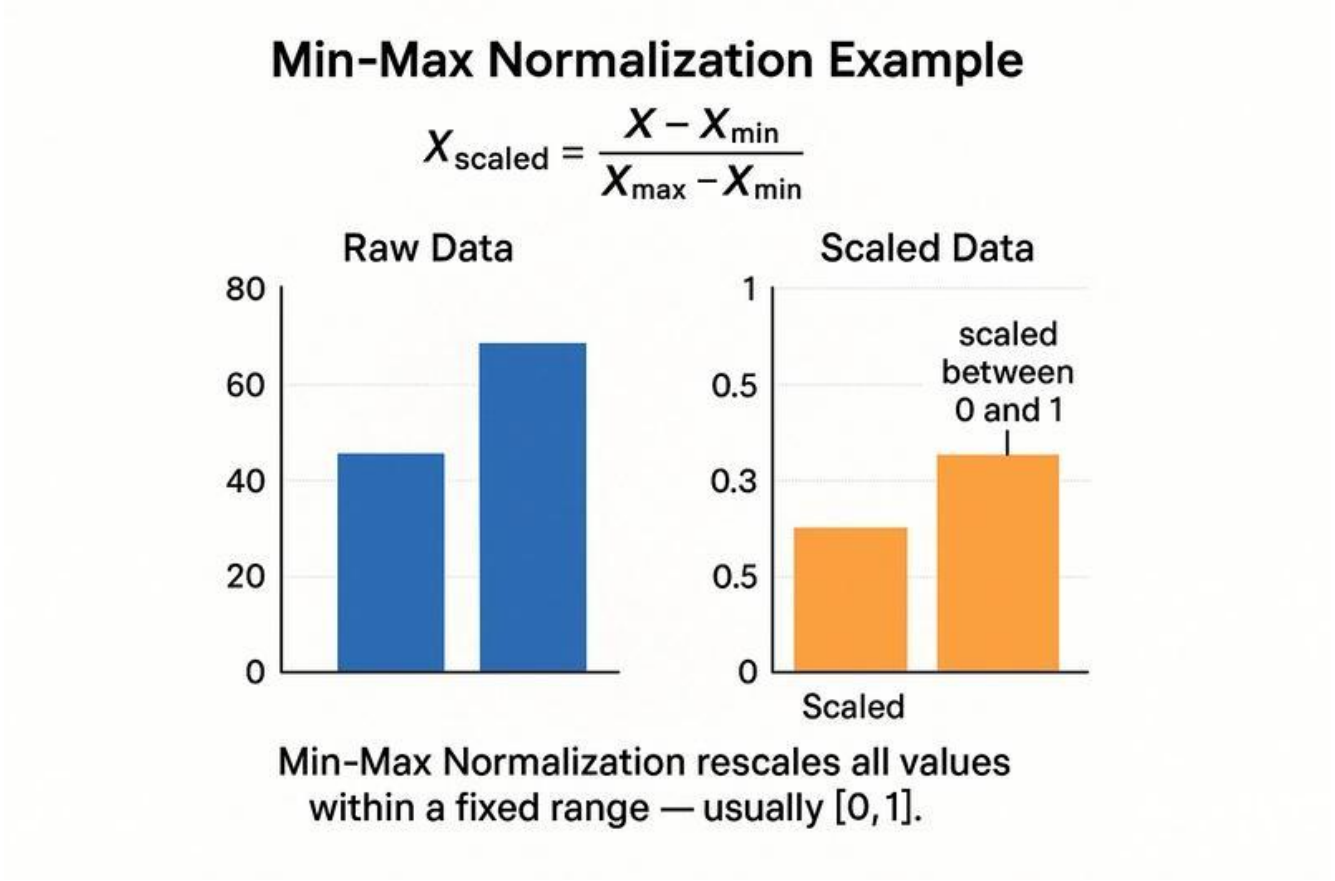
c) Importance of normalization

- Distance-based learning: Algorithms such as KNN and K-means use distance calculations. Normalization prevents large-scale features from dominating distance.
- Gradient-based learning: It helps gradient descent update model parameters more efficiently.
- Improved convergence: It reduces numerical imbalance during training.
- Better model performance: It allows the model to learn from all features more fairly.

Numerical verification: All normalized feature values are between 0 and 1.

Result: The patient data was successfully represented as a feature matrix and normalized using the Min-Max normalization technique.

Output:



TASK 2 – Vector Representation and Matrix Operations (2 Marks)

a) Represent patient features and importance weights as vectors

Consider one patient with the following normalized feature values:

Feature	**Patient value**	**Importance weight**
Age	0.60	0.10
Glucose	0.80	0.40
Blood Pressure	0.70	0.30
Cholesterol	0.50	0.20

Patient feature vector:

$$X = [0.6,\ 0.8,\ 0.7,\ 0.5]$$

Importance weight vector:

$$W = [0.1,\ 0.4,\ 0.3,\ 0.2]$$

The weighted risk score is calculated using the dot product:

$$R = X \cdot W^T$$

b) Numerical calculation

$$R = (0.6 \times 0.1) + (0.8 \times 0.4) + (0.7 \times 0.3) + (0.5 \times 0.2)$$
$$R = 0.06 + 0.32 + 0.21 + 0.10$$
$$\boxed{R = 0.69}$$

Feature	Value × Weight	Contribution
Age	0.60 × 0.10	0.06
Glucose	0.80 × 0.40	0.32
Blood Pressure	0.70 × 0.30	0.21
Cholesterol	0.50 × 0.20	0.10
Total		0.69

Interpretation: The weighted score is 0.69. Glucose contributes the highest value because it has the highest assigned importance weight.

This is an illustrative mathematical score, not a clinically validated disease-risk probability.

Result: Patient features and importance weights were represented using vectors, and the weighted score was calculated successfully.

TASK 3 – Probability and Conditional Probability (6 Marks)

Given:

- Total patients = 1000
- Patients with disease = 200
- Patients without disease = 800
- Diseased patients with elevated glucose = 160
- Non-diseased patients with elevated glucose = 120

Event definitions

- D = Patient has the disease.

- G = Patient has elevated glucose.
- D' = Patient does not have the disease.

Patient distribution table:

Patient Category	Elevated Glucose	Normal Glucose	Total
Disease	160	40	200
No Disease	120	680	800
Total	280	720	1000

a) Probability that a randomly selected patient has elevated glucose

Formula:

$$P(G) = \frac{\text{Patients with elevated glucose}}{\text{Total patients}}$$

$$P(G) = \frac{160 + 120}{1000}$$

$$P(G) = \frac{280}{1000}$$

$$\boxed{P(G) = 0.28 = 28\%}$$

b) Probability that a patient has the disease given elevated glucose

Using conditional probability:

$$P(D|G) = \frac{P(D \cap G)}{P(G)}$$

$$P(D|G) = \frac{160}{280}$$

$$\boxed{P(D|G) = 0.5714 = 57.14\%}$$

c) Probability of elevated glucose given that the patient has the disease

$$P(G|D) = \frac{P(G \cap D)}{P(D)}$$

$$P(G|D) = \frac{160}{200}$$

$$\boxed{P(G|D) = 0.8 = 80\%}$$

d) Healthcare interpretation

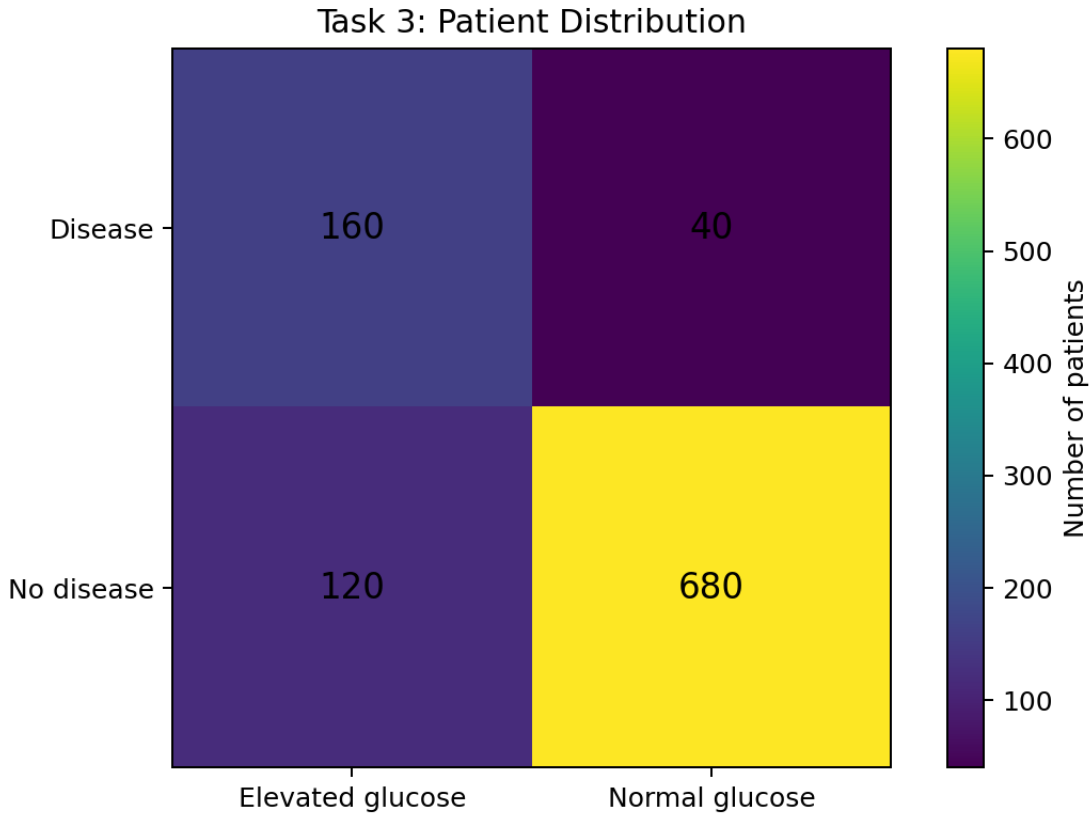
Conditional probability helps hospitals identify patients who may require additional medical testing.

- 28% of all patients have elevated glucose.

- Among patients with elevated glucose, approximately 57.14% have the disease in this sample.
- Among diseased patients, 80% have elevated glucose.

The hospital can prioritize patients with elevated glucose for further assessment. However, glucose levels alone cannot confirm a disease diagnosis.

Result: The probabilities were calculated using the total patient distribution and conditional probability formulas.
Output:



TASK 4 – Bayesian Statistics for Disease Classification (4 Marks)

Given:

- Disease prevalence = 10%
- Test positive for diseased patients = 90%
- Test positive for non-diseased patients = 5%

Therefore:

$$P(D) = 0.10$$
$$P(D') = 0.90$$

$$P(+|D) = 0.90$$
$$P(+|D') = 0.05$$

a) Calculate the probability of having the disease given a positive test result

Using Bayes' theorem:

$$P(D|+) = \frac{P(+|D)P(D)}{P(+)}$$

First, calculate the total probability of a positive result:

$$P(+) = P(+|D)P(D) + P(+|D')P(D')$$
$$= (0.90 \times 0.10) + (0.05 \times 0.90)$$
$$= 0.09 + 0.045 = 0.135$$

Substituting into Bayes' theorem:

$$P(D|+) = \frac{0.90 \times 0.10}{0.135}$$
$$= \frac{0.09}{0.135}$$
$$\boxed{P(D|+) = 0.6667 = 66.67\%}$$

b) Roles of prior probability and likelihood

Term	**Meaning**	**Value**
Prior probability	Probability of having the disease before testing	10%
Likelihood	Probability of a positive test when disease is present	90%
False-positive likelihood	Probability of a positive test without disease	5%
Posterior probability	Probability of disease after a positive test	66.67%

c) Importance of Bayesian reasoning

1. It updates disease probabilities when new test information becomes available.
2. It considers the existing prevalence of a disease.
3. It helps doctors understand uncertainty in medical test results.
4. It combines previous knowledge with new evidence.
5. It supports better decision-making when information is incomplete.

Interpretation: Even though the test has 90% sensitivity, a positive result gives a 66.67% probability of disease because the disease prevalence is only 10%.

Result: Bayes' theorem was successfully applied to calculate the posterior probability of disease.

TASK 5 – Entropy and Information Gain (3 Marks)

Given:

- Total records = 100
- High-risk patients = 40
- Low-risk patients = 60

a) Calculate entropy

Entropy measures the uncertainty or impurity present in a dataset.

Formula:

$$H(S) = -\sum_{i=1}^{n} p_i \log_2 p_i$$

Probability of high-risk patients:

$$P(H) = \frac{40}{100} = 0.4$$

Probability of low-risk patients:

$$P(L) = \frac{60}{100} = 0.6$$

Substituting:

$$H(S) = -[0.4 \log_2(0.4) + 0.6 \log_2(0.6)]$$
$$= -[0.4(-1.3219) + 0.6(-0.7370)]$$
$$= 0.5288 + 0.4422$$

$$\boxed{H(S) = 0.971 \text{ bits}}$$

b) What does entropy indicate?

- Entropy measures the uncertainty in the classification of patient records.
- Higher entropy indicates greater uncertainty.
- Lower entropy indicates that the dataset contains more records belonging to one class.
- Zero entropy means all records belong to a single class.

Here, entropy is approximately 0.971 bits, which indicates relatively high uncertainty because the two classes are reasonably balanced.

c) How information gain selects useful features

Information Gain measures how much a feature reduces uncertainty when the dataset is divided.

Formula:

$$IG(S, A) = H(S) - H(S|A)$$

Where:

- H(S) = Original entropy.
- H(S|A) = Entropy after splitting using feature A.

For example, glucose level can divide patients into groups with different risk distributions.

A feature that produces a larger information gain is generally more useful for classification.

Result: The entropy of the medical dataset was calculated as approximately 0.971 bits. Information gain can be used to select useful features such as glucose level.

TASK 6 – Numerical Computation and Distance Measurement (3 Marks)

Given:

Patient A:

$$A = (0.70, 0.80, 0.60)$$

Patient B:

$$B = (0.50, 0.75, 0.90)$$

Features: Glucose, Blood Pressure and Cholesterol.

a) Calculate Euclidean distance

Formula:

$$d(A, B) = \sqrt{\sum_{i=1}^{n} (A_i - B_i)^2}$$

Substituting the values:

$$d(A, B) = \sqrt{(0.70 - 0.50)^2}$$
$$+\sqrt{\quad}$$

More directly, expanding all three squared differences:

$$d(A, B) = \sqrt{(0.70 - 0.50)^2 + (0.80 - 0.75)^2 + (0.60 - 0.90)^2}$$
$$= \sqrt{0.04 + 0.0025 + 0.09}$$

$$= \sqrt{0.1325}$$
$$\boxed{d(A, B) = 0.364}$$

b) How distance indicates similarity

- Smaller distance indicates that two patients have more similar feature values.
- Larger distance indicates that their feature values are more different.
- Distance can be used in KNN and K-means algorithms to identify similar patient records.

Here, the distance between Patient A and Patient B is approximately 0.364.

c) Importance of normalization

Normalization is important because the features have different numerical ranges.

For example, cholesterol may have values around 200, while normalized glucose may have a value of 0.7.

Without normalization, cholesterol could dominate the distance calculation.

Result: The Euclidean distance between the two normalized patient records was calculated as approximately 0.364.

TASK 7 – Bias–Variance Trade-off (4 Marks)

Given three disease-risk models:

Model	**Training Error**	**Validation Error**
Model A	High	High
Model B	Very Low	High
Model C	Low	Low and similar to training

a) Identify high bias and high variance

Model	**Behaviour**	**Identification**
Model A	High training and validation error	High Bias
Model B	Very low training error but high validation error	High Variance
Model C	Low and similar training and validation error	Good Generalization

Model A – High Bias

The model is too simple to learn important patterns in patient data. This is called underfitting.

Model B – High Variance

The model learns the training data very closely but performs poorly on unseen patient records. This is called overfitting.

Model C – Good Generalization

The model performs consistently on both training and validation data.

b) Effect of model complexity on bias and variance

Model Complexity	**Bias**	**Variance**
Low	High	Low
Moderate	Balanced	Balanced
High	Low	High

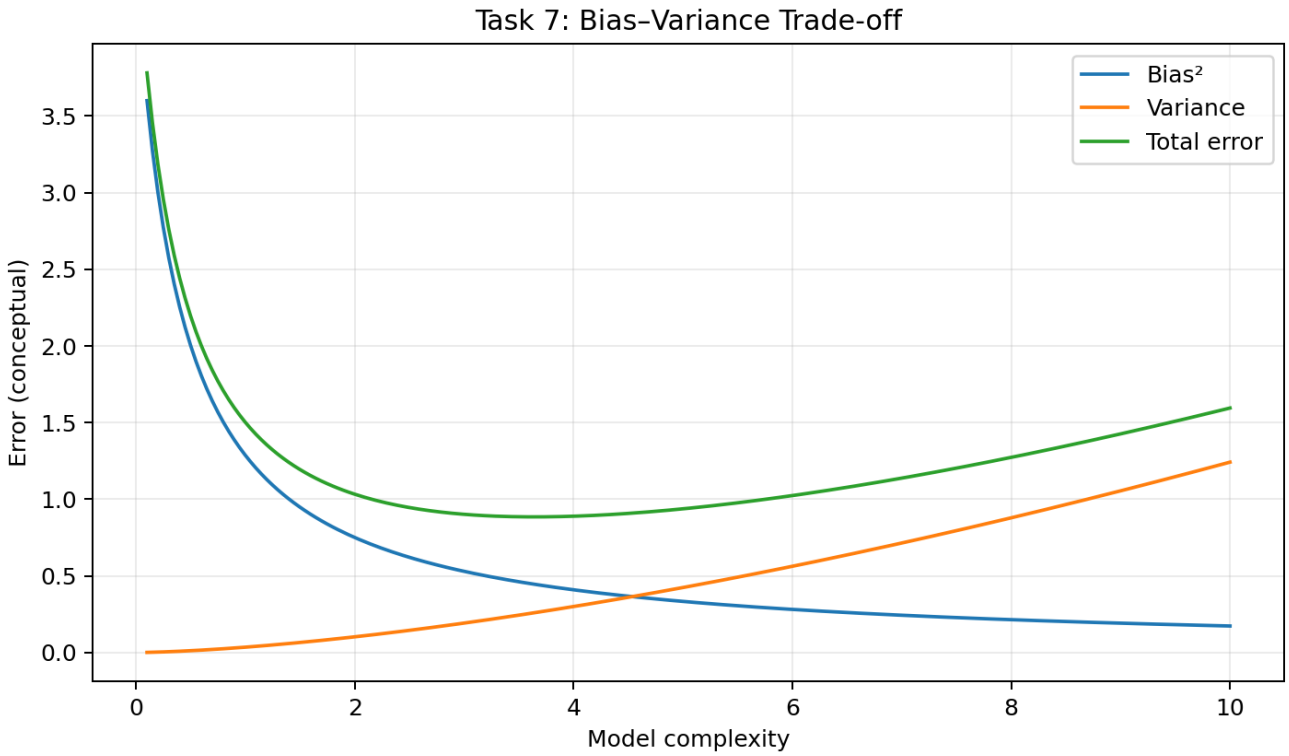
- Increasing model complexity generally decreases bias.
- Increasing model complexity generally increases variance.
- A suitable model should balance both.

c) Improvements for Models A and B

Model	**Problem**	**Suggested Improvement**
Model A	Underfitting	Increase model complexity, add useful features and train adequately.
Model B	Overfitting	Apply regularization, dropout, pruning, early stopping and more training data.

Result: Model A was identified as a high-bias model, Model B as a high-variance model and Model C as a well-generalized model.

Output Image – Bias–Variance Trade-off



TASK 8 – Overfitting, Underfitting and Model Capacity (4 Marks)

a) Identify the three model behaviours

Model	**Description**	**Behaviour**
Simple model	Fails to capture disease patterns	Underfitting
Moderately complex model	Performs consistently	Good Generalization
Highly complex model	Excellent training performance but poor unseen-data performance	Overfitting

b) How model capacity affects learning

Model capacity refers to the ability of a machine learning model to learn complex relationships from data.

- Low-capacity models learn only simple relationships.
- Moderate-capacity models can learn useful patterns without memorizing the training data.
- High-capacity models can learn complicated relationships but may also memorize noise and irrelevant patterns.

c) Why increasing capacity may improve and later reduce performance

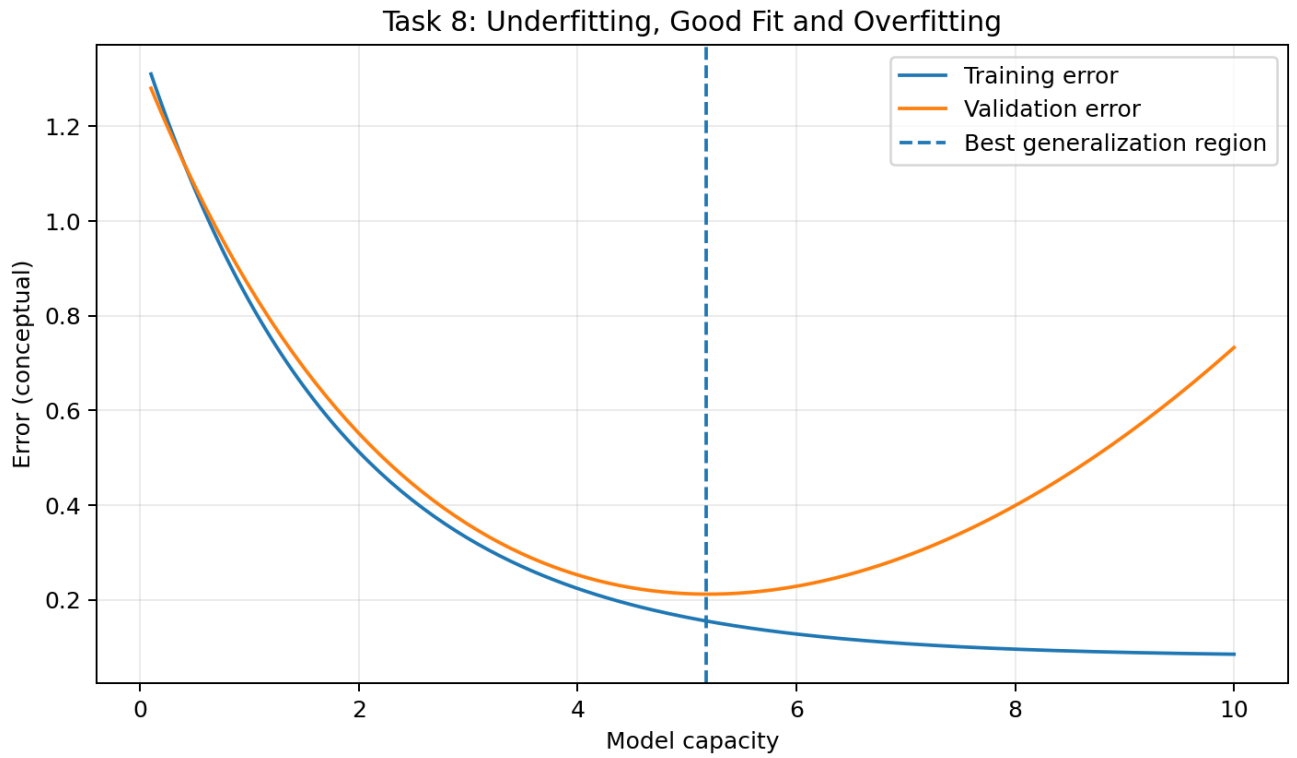
Initially, increasing model capacity helps the model learn more complex disease patterns and reduces underfitting.

However, when capacity becomes too high, the model may memorize training examples rather than learning general patterns.

This increases validation error and leads to overfitting.

Result: A suitable model capacity is required to achieve good generalization on unseen patient records.

Output Image – Model Capacity



TASK 9 – Supervised Learning and Model Evaluation (3 Marks)

a) Why is this a supervised learning problem?

The hospital has historical patient records containing input features and predefined disease labels.

For example:

Age	Glucose	Blood Pressure	Disease Label
25	90	120	No Disease
45	130	140	Disease
60	110	160	No Disease
50	150	150	Disease

The model learns the relationship between patient features and disease labels.

Therefore, this is a supervised learning problem because the training dataset contains labelled examples.

b) Roles of training and testing sets

Training Set	Testing Set
Used to train the model	Used to evaluate the model
Contains input features and labels	Contains unseen input features and labels
Helps the model learn patterns	Checks prediction performance
Used during model fitting	Used after training for final evaluation

c) Calculate accuracy

Given:

- Total test patients = 100
- Correctly classified patients = 92

Formula:

$$Accuracy = \frac{\text{Correct Predictions}}{\text{Total Predictions}} \times 100$$
$$= \frac{92}{100} \times 100$$
$$\boxed{Accuracy = 92\%}$$

Measurement	Value
Total test patients	100
Correct predictions	92
Incorrect predictions	8
Accuracy	92%

Interpretation: The model correctly classified 92 out of 100 test patients. Therefore, the model achieved 92% accuracy on the test dataset.

Accuracy alone does not show false-positive and false-negative rates, so additional evaluation metrics may be required.

Result: Supervised learning was identified as the appropriate approach, and the model accuracy was calculated as 92%.

TASK 10 – Unsupervised Learning and Patient Grouping (4 Marks)

a) Why is unsupervised learning suitable?

The hospital has patient records containing age, blood pressure, glucose and cholesterol but does not have predefined disease labels.

Unsupervised learning is suitable because it identifies hidden patterns and groups similar patient records without requiring labelled outputs.

b) How K-means groups patients

K-means is an unsupervised clustering algorithm that divides data into K clusters based on similarity.

Steps involved:

1. Select the number of clusters (K).
2. Initialize K cluster centroids.

3. Calculate the distance between each patient and the centroids.
4. Assign each patient to the nearest centroid.
5. Recalculate the centroids.
6. Repeat until the clusters become stable.

Example: Assume the hospital selects K = 3.

Cluster	Patient Characteristics	Possible Interpretation
Cluster 1	Lower glucose and blood pressure values	Group requiring routine assessment
Cluster 2	Moderate glucose and blood pressure values	Group requiring further observation
Cluster 3	Higher glucose and blood pressure values	Group that may need further investigation

These descriptions are illustrative and do not establish clinical risk categories.

c) How distance measures similarity

K-means commonly uses Euclidean distance.

$$d(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

Patients with smaller distances are assigned to similar clusters.

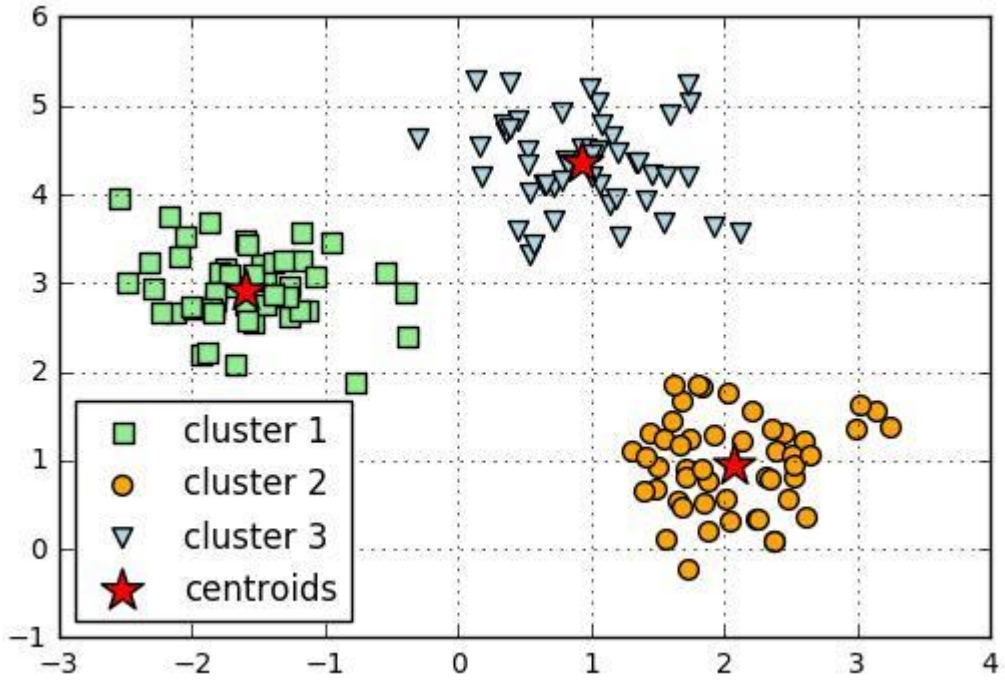
d) How clusters support clinical investigation

- Identify groups with similar patient characteristics.
- Discover patterns that may not be visible through individual records.
- Help hospitals prioritize further investigation.
- Support exploratory analysis and healthcare planning.
- Help identify unusual patient groups for additional review.

Clustering supports clinical investigation but does not replace professional medical diagnosis.

Result: K-means clustering was identified as a suitable approach for grouping unlabelled patient records based on similarity.

Output Image – K-means Patient Clustering



TASK 11 – Comparison of Supervised and Unsupervised Learning (4 Marks)

a) Identify the suitable learning paradigm

Dataset	**Data Availability**	**Suitable Learning**
Dataset A	Patient features with disease labels	Supervised Learning
Dataset B	Patient features without disease labels	Unsupervised Learning

Dataset A: Supervised learning is suitable because the model has predefined disease labels and can learn to predict disease status.

Dataset B: Unsupervised learning is suitable because the model must discover hidden patterns without predefined labels.

b) Compare objectives and outputs

Comparison	**Supervised Learning**	**Unsupervised Learning**
Input	Labelled data	Unlabelled data
Objective	Predict outcomes	Discover hidden patterns
Output	Predicted classes or values	Clusters or patterns

Algorithms	Decision Tree, SVM, Random Forest	K-means, Hierarchical Clustering
Evaluation	Accuracy, Precision, Recall, F1-score	Silhouette Score, Davies–Bouldin Index
Example	Predict disease status	Group similar patients

c) Recommend an approach for predicting disease status

Supervised learning is recommended for predicting disease status in a new patient because the hospital already has historical patient records with disease labels.

A classification algorithm such as Random Forest or Logistic Regression can learn from the labelled data and predict whether a new patient belongs to the disease or disease-free class.

Unsupervised learning can additionally be used to discover hidden patient groups.

Result: Supervised learning is recommended for disease prediction, while unsupervised learning is suitable for patient-group discovery.

TASK 12 – Design Challenge: Selecting and Adapting a Learning Strategy (8 Marks)

A. Impact Analysis (2 Marks)

a) Effect of increasing dataset size from 10,000 to 1,000,000 records

The hospital initially processes 10,000 patient records. The new requirement increases the dataset to 1,000,000 records.

This change affects data representation, numerical computation and learning efficiency.

1. Data Representation

- The feature matrix becomes significantly larger.
- Each row represents one patient.
- Each column represents a patient feature.
- Efficient data structures are required to store and process large datasets.

For example, assuming five features:

Original dataset:

$$X_{10000} \in R^{10000\times5}$$

New dataset:

$$X_{1000000} \in R^{1000000\times5}$$

2. Numerical Computation

- More calculations are required during training.
- Distance calculations become computationally expensive.
- Memory usage increases.
- Large-scale matrix operations require efficient computation.

3. Learning Efficiency

- Training time may increase.
- Batch processing and parallel computation can improve efficiency.
- Mini-batch gradient descent can process data in smaller groups.
- Efficient algorithms are needed to handle large datasets.

Comparison Table

Parameter	**Original Dataset**	**New Dataset**
Number of records	10,000	1,000,000
Number of features	5	5
Total feature values	50,000	5,000,000
Data processing	Relatively simple	More computationally demanding
Memory requirement	Lower	Higher
Training time	Lower	Potentially higher

b) Importance of normalization, missing-value handling and data variability

1. Normalization

Normalization converts features into a comparable numerical range. It prevents large-scale features from dominating distance calculations and improves learning efficiency.

2. Missing-Value Handling

Patient records may contain missing glucose, blood pressure or cholesterol values.

Possible techniques:

- Mean imputation for numerical features.

- Median imputation when outliers are present.
- Mode imputation for categorical features.
- Missing-value indicators to preserve information about missingness.

3. Data Variability

Patient records may contain different age groups, glucose levels and medical characteristics.

Understanding variability helps the model learn meaningful patterns and improves its ability to generalize to unseen patients.

Challenge	**Proposed Solution**
Different feature scales	Min-Max normalization or standardization
Missing numerical values	Median or mean imputation
Missing categorical values	Mode imputation
Large dataset	Batch and distributed processing
Different patient characteristics	Statistical analysis and feature selection

Result: Increasing the dataset size requires efficient data representation, numerical computation and data preprocessing techniques.

B. Evaluation of the Original Approach (2 Marks)

a) Is a single supervised approach sufficient for both disease prediction and patient-group discovery?

No. A single supervised learning approach is not sufficient for both tasks.

Supervised learning requires labelled data and is mainly used to predict predefined outcomes.

However, patient-group discovery requires identifying hidden patterns without predefined labels.

Requirement	**Supervised Learning**	**Unsupervised Learning**
Disease prediction	Suitable	Not directly suitable
Patient-group discovery	Limited	Suitable
Requires labels	Yes	No

Main output	Predicted disease class	Patient clusters
Example algorithm	Random Forest	K-means

Therefore, the hospital should combine supervised classification and unsupervised clustering.

b) Limitations of labelled learning when useful patterns exist in unlabelled records

A supervised model learns from labelled patient records only.

When unlabelled patient records are available, the following limitations may occur:

1. Useful patterns in unlabelled records may remain undiscovered.
2. The model may not identify new patient groups.
3. Limited labelled data may reduce model generalization.
4. Hidden relationships between patient features may be ignored.
5. The model may fail to identify unusual patient characteristics.

Therefore, combining labelled and unlabelled data provides a more comprehensive understanding of the patient population.

Result: A single supervised approach is insufficient. A combination of supervised and unsupervised learning is more appropriate.

C. Adaptation or Redesign (2 Marks)

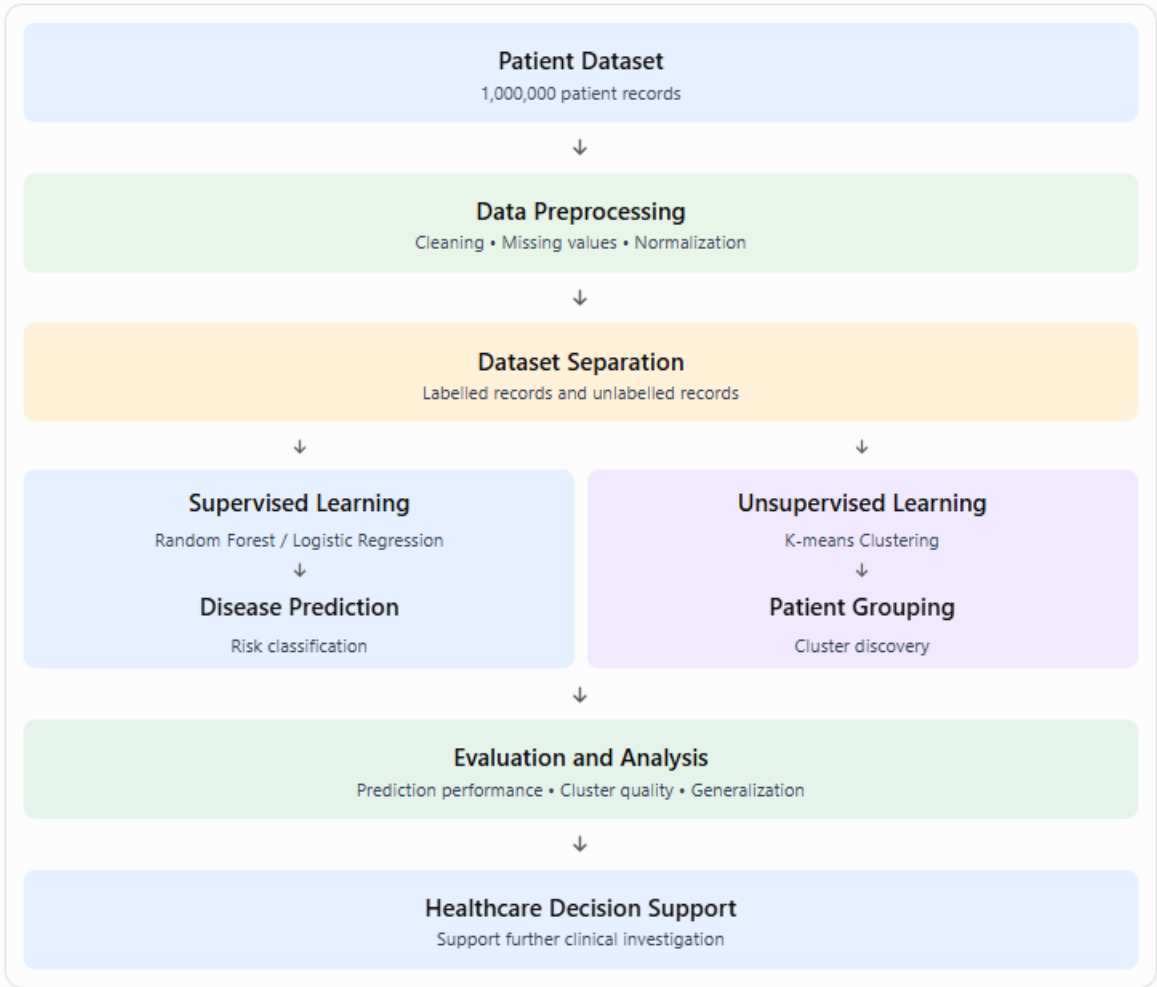
a) Proposed strategy combining supervised classification and unsupervised clustering

The hospital should redesign its original approach by implementing a hybrid learning system.

The proposed system consists of the following stages:

1. Collect 1,000,000 patient records.
2. Perform data cleaning and preprocessing.
3. Handle missing values.
4. Normalize numerical features.
5. Separate labelled and unlabelled records.
6. Apply supervised classification to predict disease risk.
7. Apply K-means clustering to discover patient groups.
8. Evaluate both models and generate results.

Proposed Hybrid Learning Architecture



b) Incorporating probability, distance measures, normalization and model capacity

1. Probability

Probability can estimate disease likelihood based on patient features. Bayesian reasoning can update the probability when new medical information becomes available.

2. Distance Measures

Euclidean distance can measure similarity between normalized patient records. K-means uses distance to assign patients to clusters.

3. Normalization

Min-Max normalization or standardization can bring numerical features into comparable ranges before applying distance-based algorithms.

4. Model Capacity

Model capacity should be selected carefully to avoid underfitting and overfitting.

Regularization, cross-validation and early stopping can improve generalization.

Mathematical Concept	Application in Proposed System
Probability	Estimate disease likelihood
Conditional Probability	Analyze relationships between medical features
Euclidean Distance	Measure patient similarity
Normalization	Balance feature scales
Model Capacity	Control model complexity
Regularization	Reduce overfitting

Result: A hybrid learning strategy combining classification and clustering is proposed to support disease prediction and patient-group discovery.

D. Evidence-Based Defence (2 Marks)

a) Suitable experiments and evaluation measures

To compare the original and redesigned approaches, the hospital should conduct the following experiments.

Experiment 1 – Dataset Scaling

Train the model using 10,000, 100,000 and 1,000,000 records.

Measure training time, memory usage and prediction performance.

Experiment 2 – Normalization Comparison

Compare model performance using raw patient data and normalized patient data.

Experiment 3 – Missing-Value Handling

Compare different imputation methods such as mean and median imputation.

Experiment 4 – Learning Strategy Comparison

Compare a supervised-only approach with the proposed hybrid learning approach.

Experiment 5 – Clustering Evaluation

Apply K-means with different values of K and evaluate cluster quality.

Evaluation Metrics Table

Evaluation Metric	Purpose
Accuracy	Measures correct disease predictions
Precision	Measures correctness of positive predictions
Recall	Measures the ability to identify diseased patients
F1-Score	Balances precision and recall
ROC-AUC	Measures classification discrimination
Silhouette Score	Measures cluster separation and similarity
Davies–Bouldin Index	Measures cluster compactness and separation
Training Time	Measures computational efficiency
Memory Usage	Measures resource requirements

b) Justification of the final design

The hybrid learning approach is selected because it satisfies both disease prediction and patient-group discovery requirements.

Reasons for selecting the redesigned approach:

1. Mathematical Reasoning: Probability, vector operations and distance measures provide a mathematical foundation for prediction and clustering.
2. Generalization: Regularization and validation help prevent overfitting.
3. Computational Requirements: Mini-batch processing and efficient matrix operations help manage one million records.
4. Data Characteristics: Missing-value handling and normalization improve data quality.
5. Learning Strategy: Supervised learning predicts disease status, while unsupervised learning identifies hidden patient groups.

Final Comparison

Parameter	Original Approach	Redesigned Approach
Dataset size	10,000 records	1,000,000 records

Learning method	Single supervised model	Hybrid learning
Disease prediction	Supported	Supported
Patient grouping	Not directly supported	Supported
Missing-value handling	Basic preprocessing	Dedicated imputation
Normalization	Basic	Systematic feature scaling
Scalability	Limited for new requirements	Designed for larger datasets
Evaluation	Prediction metrics	Prediction and clustering metrics

Final Result: The original analytical strategy should be redesigned into a hybrid learning system combining supervised classification and unsupervised clustering. The redesigned approach can support disease-risk prediction, patient-group discovery and large-scale patient data analysis while maintaining appropriate model evaluation and computational efficiency.

Final Mark Distribution – CO1

Task	Topics Covered	Marks
Task 1	Feature Representation and Normalization Linear Algebra	5
Task 2	Vector Representation and Matrix Operations	2
Task 3	Probability and Conditional Probability	6
Task 4	Bayes' Theorem and Bayesian Classification	4
Task 5	Entropy and Information Gain	3
Task 6	Numerical Computation and Distance Measurement	3
Task 7	Bias Variance Trade-off	4
Task 8	Overfitting, Underfitting and Model Capacity	4
Task 9	Supervised Learning Model Evaluation	3
Task 10	Unsupervised Learning Clustering Analysis	4
Task 11	Comparison of Learning Approaches	4
Sub Total		42
Task 12	Design Challenge Selecting and Adapting a Learning Strategy	8
TOTAL		50